

Sakshi Sharma

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EDUCATION

Boston University Boston, MA Sep 2025 - Present
PhD in Computer Science : CGPA - 3.9/4
Advisor: Prof. Vasiliki Kalavri, Prof. John Liagouris
Boston University Boston, MA Sep 2022 - May 2024
MS in Computer Science : CGPA - 3.9/4
International Institute of Information Technology Bhubaneswar, India Aug 2017 - May 2021
B.Tech in Computer Science Engineering : CGPA - 8.6/10
Relevant Coursework
Streaming Systems, Advanced Databases, Distributed Systems, Programming Language Foundations for Concurrency, Programming Massively Parallel Multiprocessors and Heterogeneous Systems

PUBLICATIONS

- D. Sondhi, M. Jobanputra, D. Rani, S. Purandare, **S. Sharma** and R. Purandare, "Mining Similar Methods for Test Adaptation", *IEEE Transaction on Software Engineering*, doi: 10.1109/TSE.2021.3057163. [Link](#)
- L. Huang, **S. Sharma**, C. Yu, Z. Wang, Y. Wang, Z. Zhu, J. Liagouris, V. Kalavri, "Koala: Non-disruptive Reconfiguration for Long-running Stateful Dataflows" : Accepted Publication, NSDI 2027

PROJECTS

A Disaggregated Stream Processing System for Heterogeneous Environments Jan - Sep 2025, Boston University
Developing a fully-disaggregated design for data stream processing system that decouples core data-plane and control-plane services, enabling nonblocking online reconfiguration without pausing the computation.

Dynamic Parameter Optimization for LSM-Based Databases Masters Thesis, Boston University
Developing a dynamic parameter optimization framework for LSM-based databases in C++ , with a focus on identifying essential parameters and dynamically tuning them based on workload variations to achieve significant performance improvements. [Link](#)

Addressing transient workload spikes with Cloud bursting in Apache Flink Boston University
Developing an adaptive Flink application in Java to leverage cloud bursting technique as an alternative to back-pressure. [Link](#)

Enhancing Raft consensus algorithm to tolerate partial network connectivity Boston University
Enhancing Raft consensus algorithm to tolerate partial network failures by incorporating ballot leader election and quorum connectivity features in Go [Link](#)

WORK EXPERIENCE

Institute for Global Sustainability Boston, MA May 2023 - Dec 2024
Software Development Engineer Intern
Designed a scalable MongoDB database for the Corporate Carbon Risk project and deployed Flask APIs on Amazon EC2 to automate real-time data collection, enhancing scalability.

BNY Mellon Pune, India Feb 2021 - Aug 2022
Software Development Engineer
Detected and fixed a load balancing bug for 10 microservices, reducing API latency from 400 ms to 10 ms, and played a key role in migrating 6 Spring Boot microservices to a Uniform Product URL, ensuring seamless functionality.

Dell Technologies Hyderabad, India May 2020 - July 2020
Software Development Engineer Intern
Automated testing in Spring Boot, replicating complex configurations and mocking artifacts in Dell IT systems, reducing development time by 20%.

IIIT Delhi Delhi, India May 2019 - Aug 2019
Research Intern
Developed an NLP model and a test mining tool, revealing 67 inter-language defects across 12 GitHub libraries by detecting functionally similar code in Java and Python.